

of the locking mechanism from the plastic itself. Electrical contacts hardly offer any resistance. So, if you are into design or testing firmware development, and need to do lot of debugging, and need to connect-disconnect a connector too often, then search for the interlocking parts on the connector, take a file and simply chop off those interlocking areas so that your connector moves in and out smoothly. That's what you can do while designing."

So, if the connector manufacturers (of PCB-to-wire or pluggable-to-terminals) can create a lock-release mechanism, where if you press an area then the mechanical interlocking becomes disabled or ineffective, the force required to plug out the connector will get reduced. Take for example the Ethernet cable, which has one such mechanism. Its lock pin has been built in such a way that the connector comes out smoothly, completely breaking the connection.

A few more design issues

"Another problem is that of the 9-pin D-Sub connector or the 5-pin D-Sub connector being replaced by HDMI cables," says Mahendra Karandikar. "These cables have two screws on both sides. You put the pin and tighten the screws. What happens is that when the counter part of the screw, which is on the PCB, is unscrewed, then the counterpart also starts rotating. This leads to the counterpart, which is a nut, also coming out along with the screw. On one side the screw comes out clearly and on the other side the screw comes out with the nut. To avoid that, we apply some kind of a fastener adhesive and insert that particular nut inside the 9-pin."

Dust problems are also a cause of concern. If a PCB is not used for a long time, then the connectors stop working properly as some connectors are sensitive to dust. "I faced an issue with the RJ 45 connector. Because it was of poor quality, due to dust accumulation it stopped responding properly and we had to change the connector/wire on the whole," says Abhinav Behl.

An attractive option: Wireless connectivity

A lot of industrial OEMs incur huge cost in wiring. When you are setting

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up a factory, one of the major cost drivers is the wiring. Being able to cut that down is a huge benefit to many companies. According to Rajanikant Mohan, wireless connectivity can reduce the clutter of cables considerably.

However, it is important to also see the dangers and inconveniences that come with wireless connectivity. For instance, there can be a problem of interference, that is several wireless devices talking to each other, resulting in a clash. Also, clearing the test of RF susceptibility and immunity can cause issues.

But there is a tradeoff since your wiring is reduced with wireless connectivity. The EFTA requirements and the surge which comes through the cable is not required to be taken care of, because you do not have any cable coming in.

"If the industry goes wireless, then there are too many standards and methods of communication such as TCP-IP over Wi-Fi, Bluetooth, LoRa, Zigbee, and many more. And then there is the problem of the wireless

signal passing through when direct line of sight is not available while devices are talking to one another. Then you have to install repeaters. But the moment you install repeaters, the signal is repeated and then you have the issue of bandwidth, because the amount of data transferred through wired cable is immense. If you want to take it on wireless, where there will be master-slave kind of communication or multi-node kind of communication, then it is going to be challenging," says Mahendra Karandikar.

"What industry requires is the presence of very few kinds of communication standards and protocols. These should be laid down as per the precise requirements of the industry without any compromise. A lot of thinking needs to be done. If there are few standards, then the industry will accept it (wireless) quickly," he says.

The good news is that some problems concerning wireless have been addressed. For instance, Bluetooth has a built-in dynamic frequency hopping as part of the protocol, although it is 2.4GHz. So, where you have Wi-Fi co-existing with Bluetooth, the in-built frequency hopping helps in improving the reliability. From an encryption point of view, the security aspect is pretty much covered. There is 128-bit AES encryption which gets built-in to the Bluetooth standard nowadays.

"Wi-Fi and Bluetooth combination is really good. You have Bluetooth for some lower speed and low-energy transmissions and Wi-Fi for higher-bandwidth stuff. I think we need to start seeing more packages with Wi-Fi and Bluetooth integrated. If we come up with an integrated solution with both Bluetooth and Wi-Fi, then that will be a flexible, all-rounded solution," says Abhishek Satish.

Abhinav Behl concludes with the thought that, in the near future, there needs to be an improvement to the delay experienced in Bluetooth connectivity. Thanks to the powerful capabilities that the upcoming chipsets offer, a solution can surely be achieved soon. **EFY**

This article is based on a panel discussion on 'Connectors: What's New For Design Engineers?' during the November edition of India Technology Week